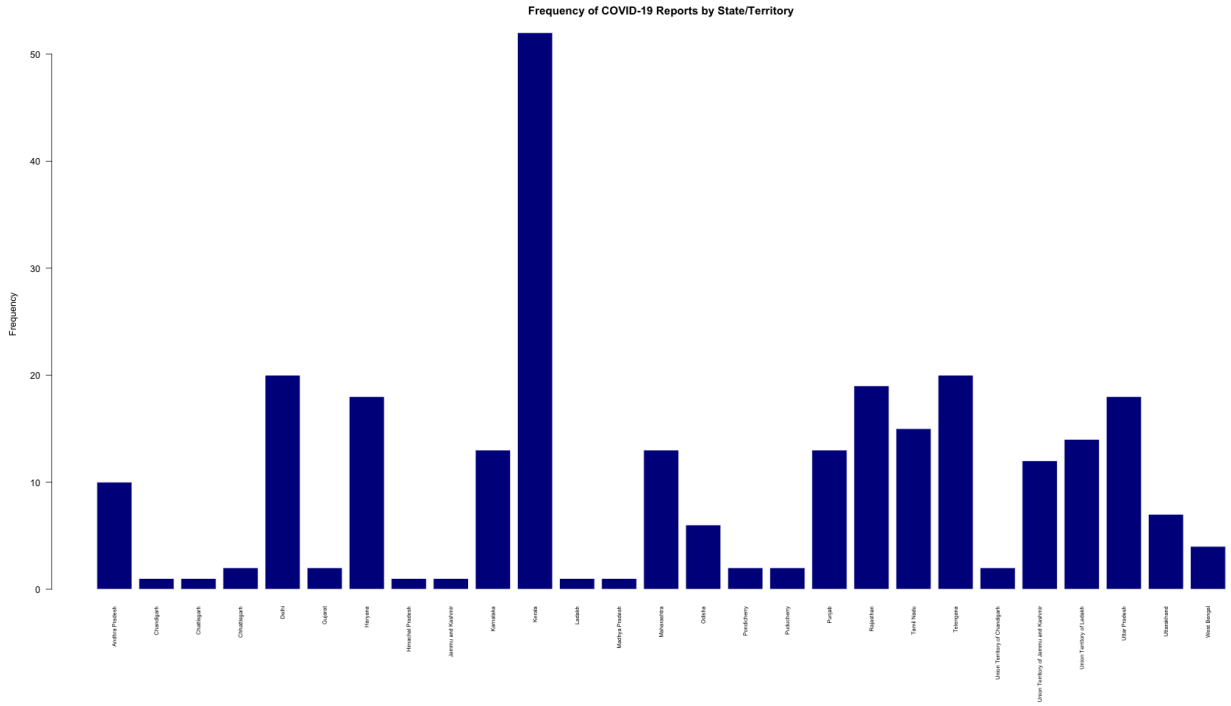
Data Activity 4: COVID-19 Data Visualisation in R

```
> # Upload file
> covid19cases <- read.csv(file.choose())
>
> # Data set structure
> str(covid19cases)
'data.frame':270 obs. of  7 variables:
 $ Sno                     : int  1 2 3 4 5 6 7 8 9 10 ...
 $ Date                    : chr  "30-01-2020" "31-01-2020" "01-02-2020" "02-02-2020" ...
 $ State.UnionTerritory    : chr  "Kerala" "Kerala" "Kerala" "Kerala" ...
 $ ConfirmedIndianNational : int  1 1 2 3 3 3 3 3 3 3 ...
 $ ConfirmedForeignNational: int  0 0 0 0 0 0 0 0 0 0 ...
 $ Cured                   : int  0 0 0 0 0 0 0 0 0 0 ...
 $ Deaths                  : int  0 0 0 0 0 0 0 0 0 0 ...
>
> # Corrects date format as it initially appears as a character format
> covid19cases$Date <- as.Date(covid19cases$Date, format = "%d-%m-%Y")
>
>
> #Task 1 Create a bar chart showing the frequency of COVID-19 reports by state/union
territory.
>
> # Create a frequency table of state/territory reports.
> frequency_table <- table(covid19cases$State.UnionTerritory)
>
> # Display a frequency table of state/territory reports.
> frequency_table

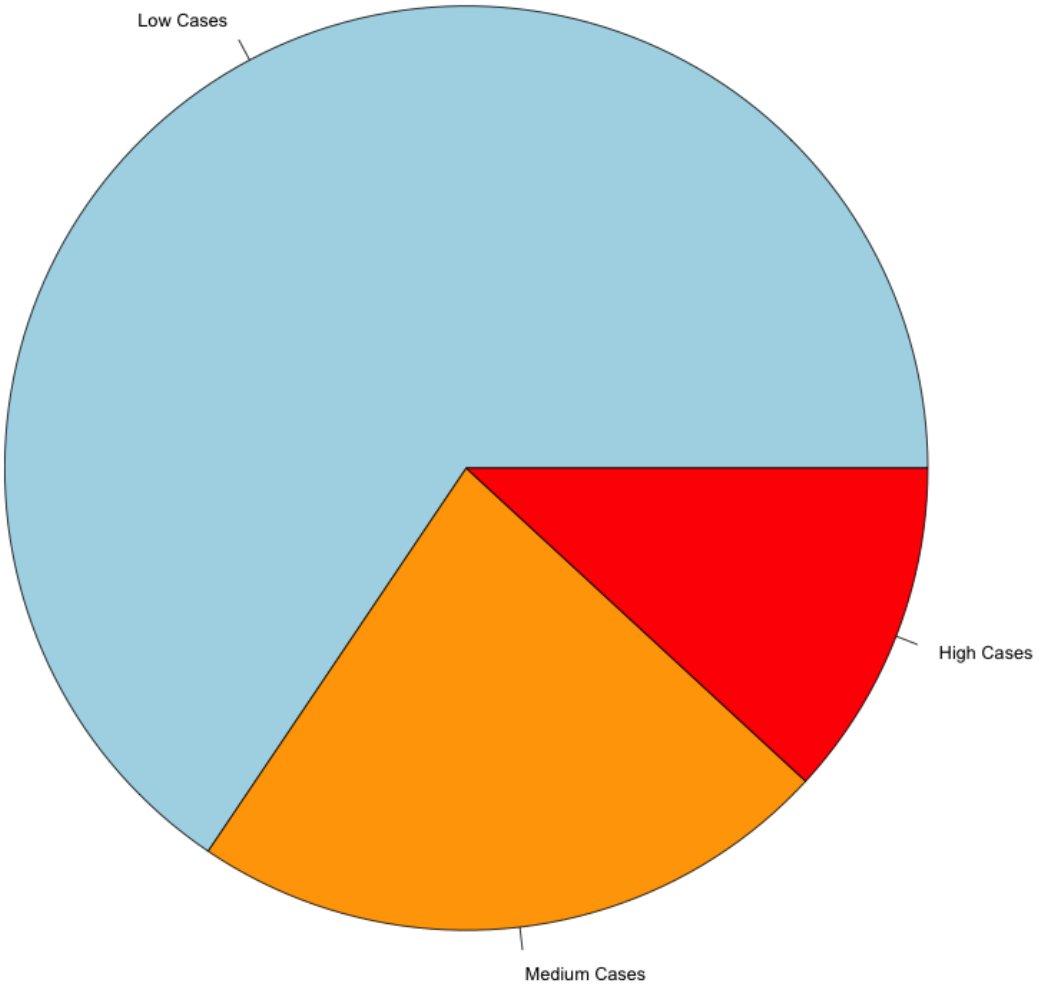
                      Andhra Pradesh                               Chandigarh
Chattisgarh                          Chhattisgarh                                    Delhi
Gujarat                               Haryana
                                    10                                        1
1                                     2                                        20
2                                    18
                    Himachal Pradesh                      Jammu and Kashmir
Karnataka                                Kerala                                    Ladakh
Madhya Pradesh                           Maharashtra
                                    1                                         1
13                                    52                                        1
1                                    13
                                Odisha                               Pondicherry
Puducherry                                Punjab                                  Rajasthan
Tamil Nadu                                Telengana
                                    6                                         2
2                                    13                                        19
15                                   20
       Union Territory of Chandigarh Union Territory of Jammu and Kashmir
Union Territory of Ladakh                          Uttar Pradesh
Uttarakhand                           West Bengal
                                    2                                        12
14                                   18                                        7
4
>
> # Increase bottom margin for vertical labels
> par(mar = c(12, 4, 4, 2))
>
> # Bar chart
> barplot(frequency_table,
+         main = "Frequency of COVID-19 Reports by State/Territory",
+         xlab = "",
+         ylab = "Frequency",
+         col = "darkblue",
+         border = "white",
+         las = 2,
+         cex.names = 0.6)
```



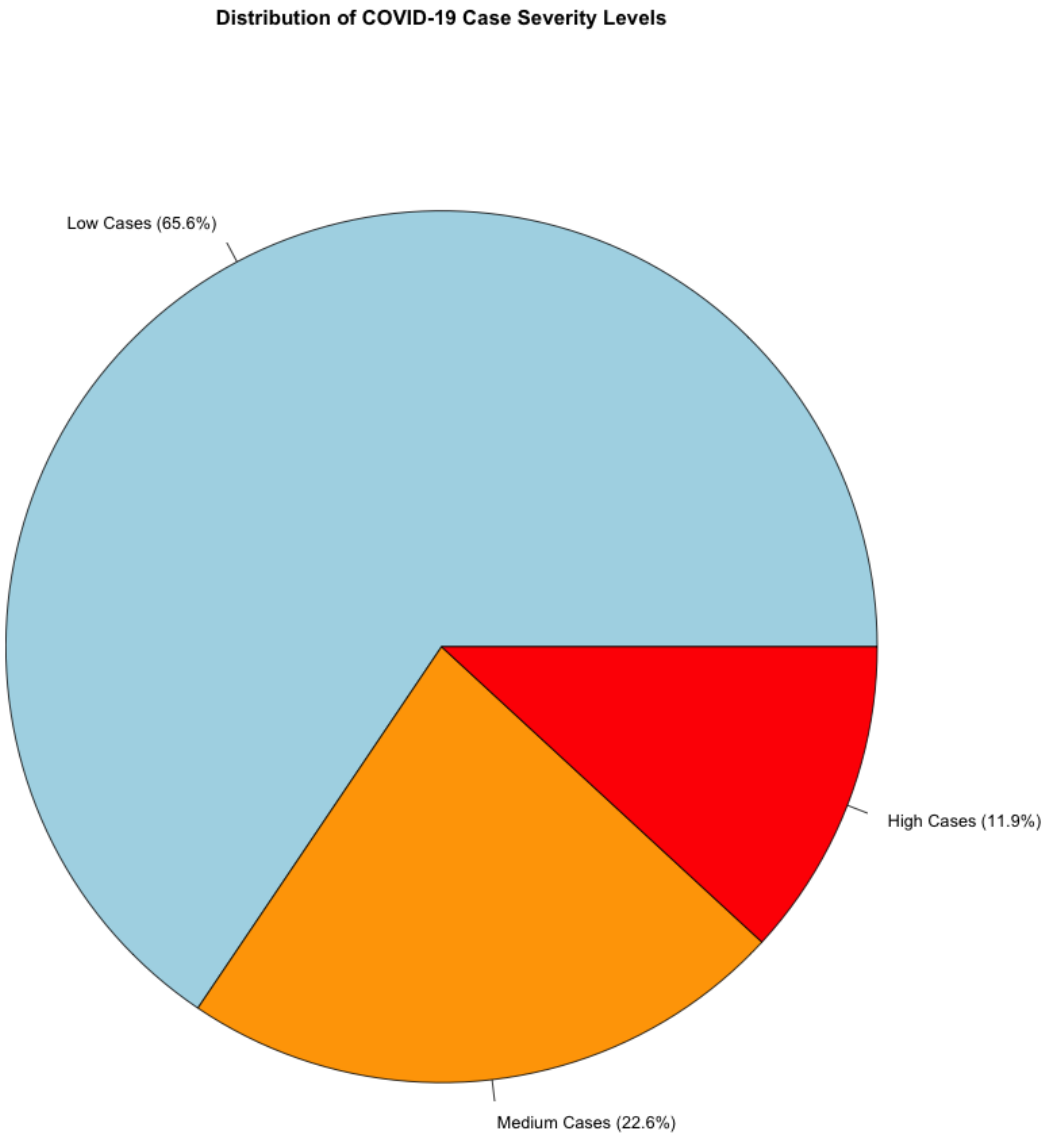

```
> # Task 2 Using variables ConfirmedIndianNational and ConfirmedForeignNational, create a
pie chart showing the distribution of case severity levels based on total confirmed cases
>
> # Total Reported Cases
> covid19cases$TotalConfirmed <- covid19cases$ConfirmedIndianNational +
+   covid19cases$ConfirmedForeignNational
>
> # Categorical Variable breaking severity up into groups
> covid19cases$case_level <- cut(
+   covid19cases$TotalConfirmed,
+   breaks = c(-Inf, 0, 5, 15, Inf),
+   labels = c("No Cases", "Low Cases", "Medium Cases", "High Cases"),
+   right = TRUE)
>
> # Create Severity Table
> case_level <- table(covid19cases$case_level)
>
> # Remove categories with zero frequency to tidy up the display
> case_level <- case_level[case_level > 0]
>
> # Reset plotting margins from Task 1 to make the chart display clearer
> par(mar = c(5, 4, 4, 2))
>
> # Pie Chart
> pie(case_level,
+     main = "Distribution of COVID-19 Case Severity Levels",
+     col = c("lightblue", "orange", "red"))
```


```
> # Task 2 Using variables ConfirmedIndianNational and ConfirmedForeignNational, create a
pie chart showing the distribution of case severity levels based on total confirmed cases
>
> # Total Reported Cases
> covid19cases$TotalConfirmed <- covid19cases$ConfirmedIndianNational +
+   covid19cases$ConfirmedForeignNational
>
```

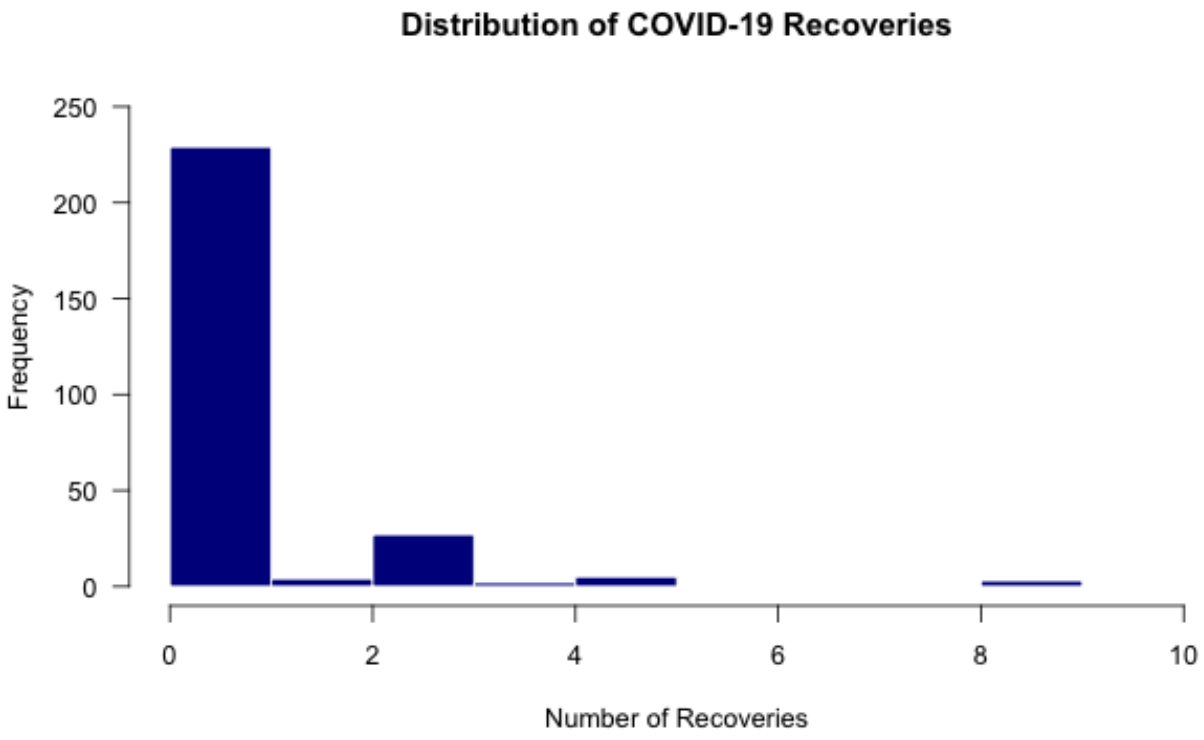
```
> # Categorical Variable breaking severity up into groups
> covid19cases$case_level <- cut(
+   covid19cases$TotalConfirmed,
+   breaks = c(-Inf, 0, 5, 15, Inf),
+   labels = c("No Cases", "Low Cases", "Medium Cases", "High Cases"),
+   right = TRUE)
>
> # Create Severity Table
> case_level <- table(covid19cases$case_level)
>
> # Remove categories with zero frequency to tidy up the display
> case_level <- case_level[case_level > 0]
>
> # Reset plotting margins from Task 1 to make the chart display clearer
> par(mar = c(5, 4, 4, 2))
>
> # Pie Chart
> pie(case_level,
+     main = "Distribution of COVID-19 Case Severity Levels",
+     col = c("lightblue", "orange", "red"))
```



```
# Add percentages to make the pie chart visually cleaner
> # Calculate percentages
> percentages <- round(case_level / sum(case_level) * 100, 1)
>
> # Create labels with percentages
> pie_labels <- paste0(names(case_level), " (", percentages, "%)")
>
> # Pie Chart with percentages
> pie(case_level,
+     labels = pie_labels,
+     main = "Distribution of COVID-19 Case Severity Levels",
+     col = c("lightblue", "orange", "red"))
```

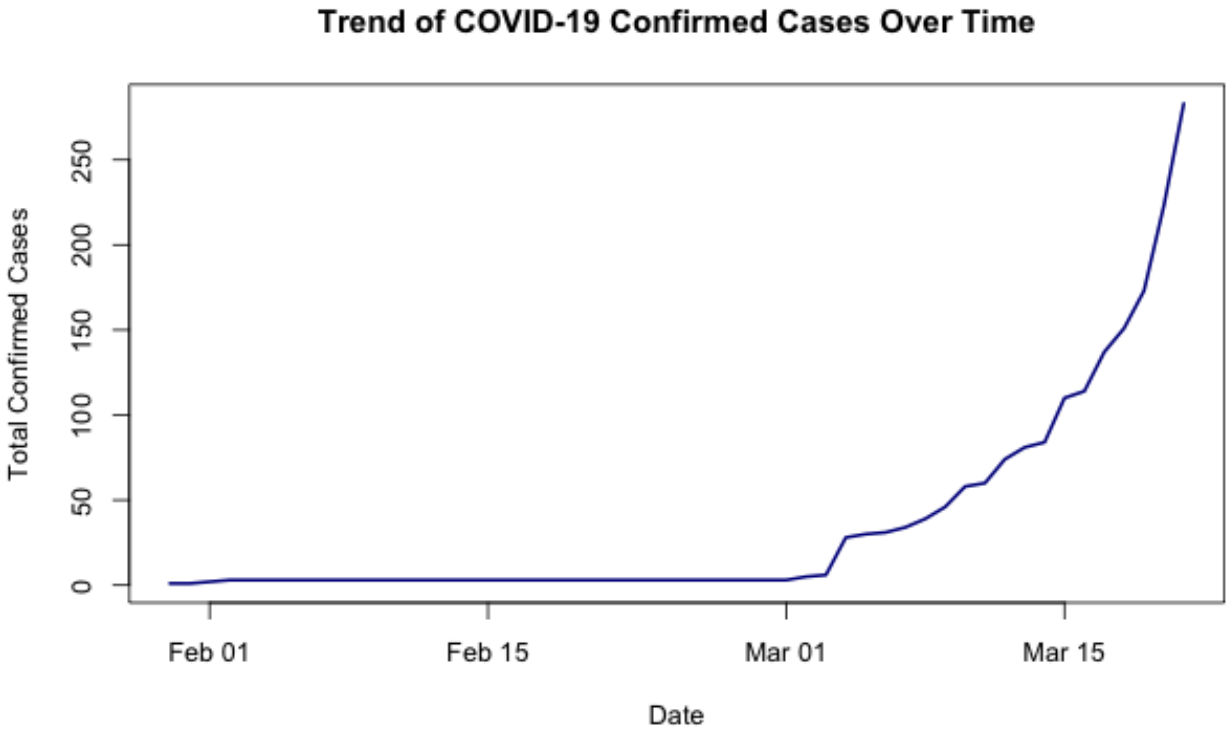


```
> #Task 3 Create a histogram showing the distribution of recovery numbers.
>
> # Create histogram
> hist(covid19cases$Cured,
+      main = "Distribution of COVID-19 Recoveries",
+      xlab = "Number of Recoveries",
+      ylab = "Frequency",
+      col = "darkblue",
+      border = "white",
+      xlim = c(0, 10),
+      ylim = c(0, 250),
+      xaxt = "n",
+      yaxt = "n")
>
> # X-axis: 0 to 10, increments of 2
> axis(1, at = seq(0, 10, by = 2))
>
> # Y-axis: 0 to 250, increments of 50
> axis(2, at = seq(0, 250, by = 50), las = 1)
```



```
> # Task 4 Create a line chart showing the trend of total cases over time.
> # Convert Date variable to date format
> covid19cases$Date <- as.Date(covid19cases$Date,
+                              format = "%d-%m-%Y")
>
> # Calculate total confirmed cases
> covid19cases$TotalConfirmed <- covid19cases$ConfirmedIndianNational +
+   covid19cases$ConfirmedForeignNational
>
> # Calculate total confirmed cases for each date
> daily_totals <- aggregate(TotalConfirmed ~ Date,
+                           data = covid19cases,
+                           FUN = sum)
>
> # Display results
> daily_totals
         Date TotalConfirmed
1  2020-01-30              1
2  2020-01-31              1
3  2020-02-01              2
4  2020-02-02              3
5  2020-02-03              3
6  2020-02-04              3
7  2020-02-05              3
8  2020-02-06              3
9  2020-02-07              3
10 2020-02-08              3
11 2020-02-09              3
12 2020-02-10              3
13 2020-02-11              3
14 2020-02-12              3
15 2020-02-13              3
16 2020-02-14              3
17 2020-02-15              3
18 2020-02-16              3
19 2020-02-17              3
20 2020-02-18              3
21 2020-02-19              3
22 2020-02-20              3
23 2020-02-21              3
24 2020-02-22              3
25 2020-02-23              3
26 2020-02-24              3
27 2020-02-25              3
28 2020-02-26              3
29 2020-02-27              3
30 2020-02-28              3
31 2020-02-29              3
32 2020-03-01              3
33 2020-03-02              5
34 2020-03-03              6
35 2020-03-04             28
36 2020-03-05             30
37 2020-03-06             31
38 2020-03-07             34
39 2020-03-08             39
40 2020-03-09             46
41 2020-03-10             58
42 2020-03-11             60
43 2020-03-12             74
44 2020-03-13             81
45 2020-03-14             84
46 2020-03-15            110
47 2020-03-16            114
48 2020-03-17            137
49 2020-03-18            151
50 2020-03-19            173
51 2020-03-20            223
52 2020-03-21            283
>
> # Sort observations chronologically
> daily_totals <- daily_totals[order(daily_totals$Date), ]
```

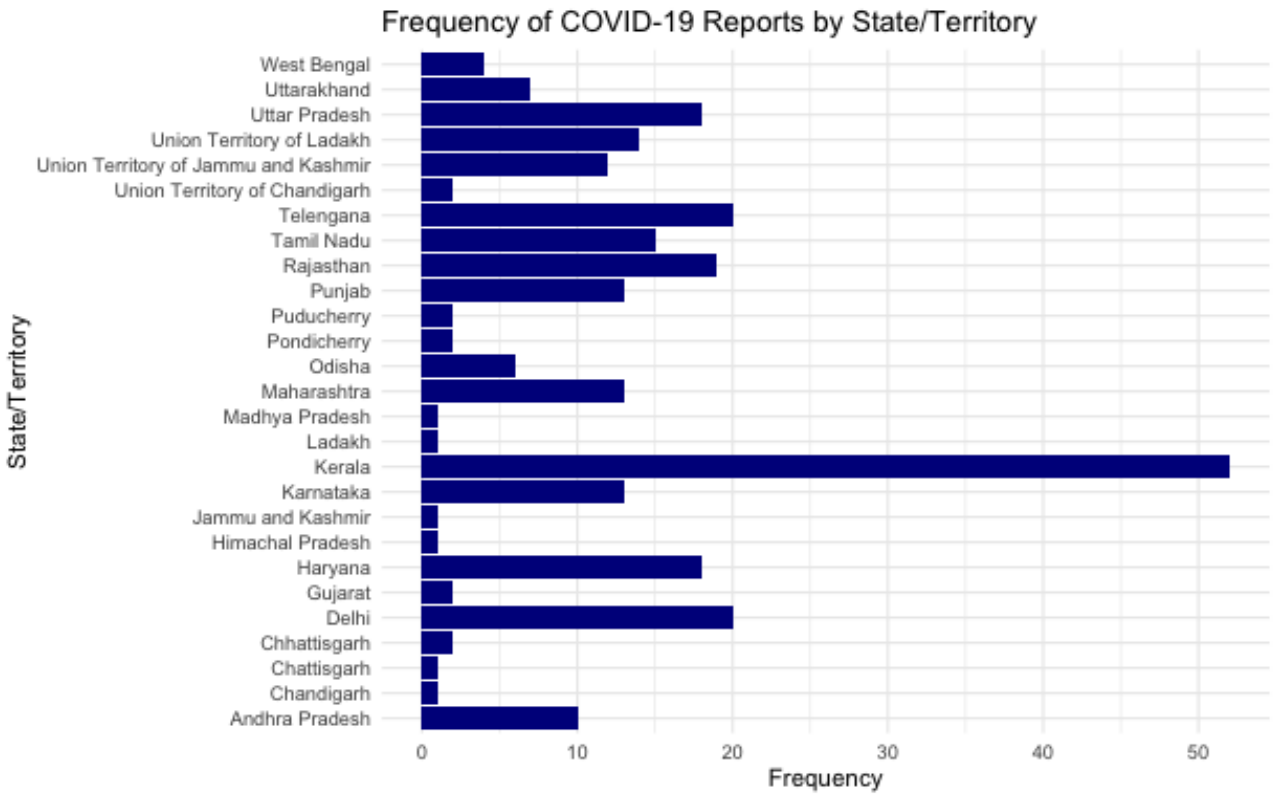
```
>
> # Create a line chart showing the trend of total cases over time
>
> plot(daily_totals$Date,
+      daily_totals$TotalConfirmed,
+      type = "l",
+      main = "Trend of COVID-19 Confirmed Cases Over Time",
+      xlab = "Date",
+      ylab = "Total Confirmed Cases",
+      col = "darkblue",
+      lwd = 2)
```



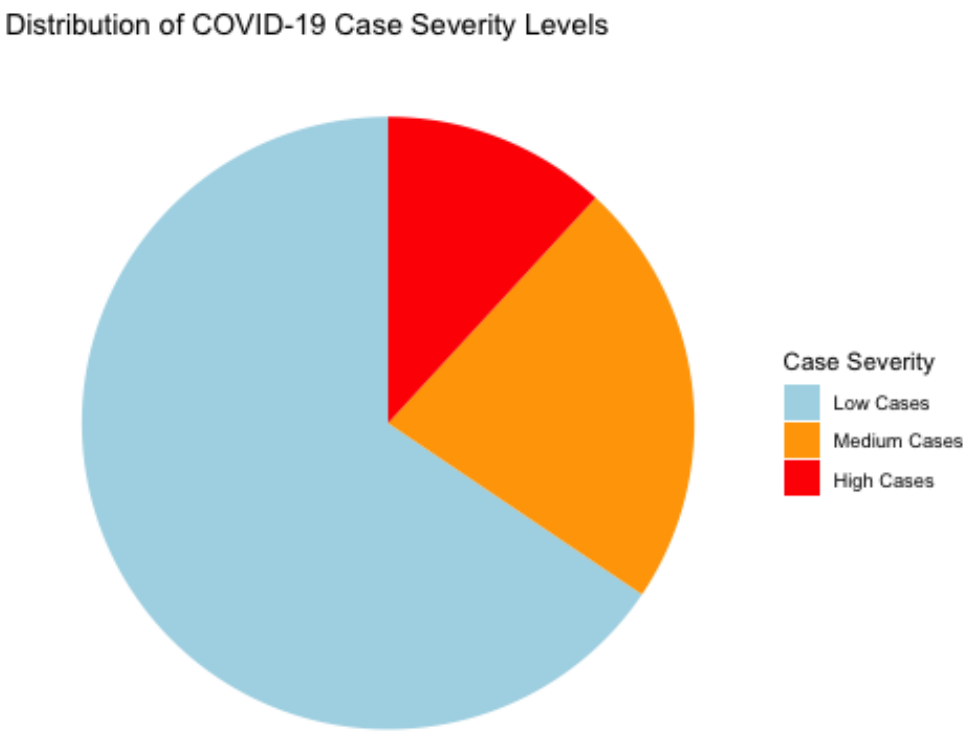
Repeat using GGPLOT

```
# Task 1 Create a bar chart showing the frequency of COVID-19 reports by state/union
territory.
library(ggplot2)

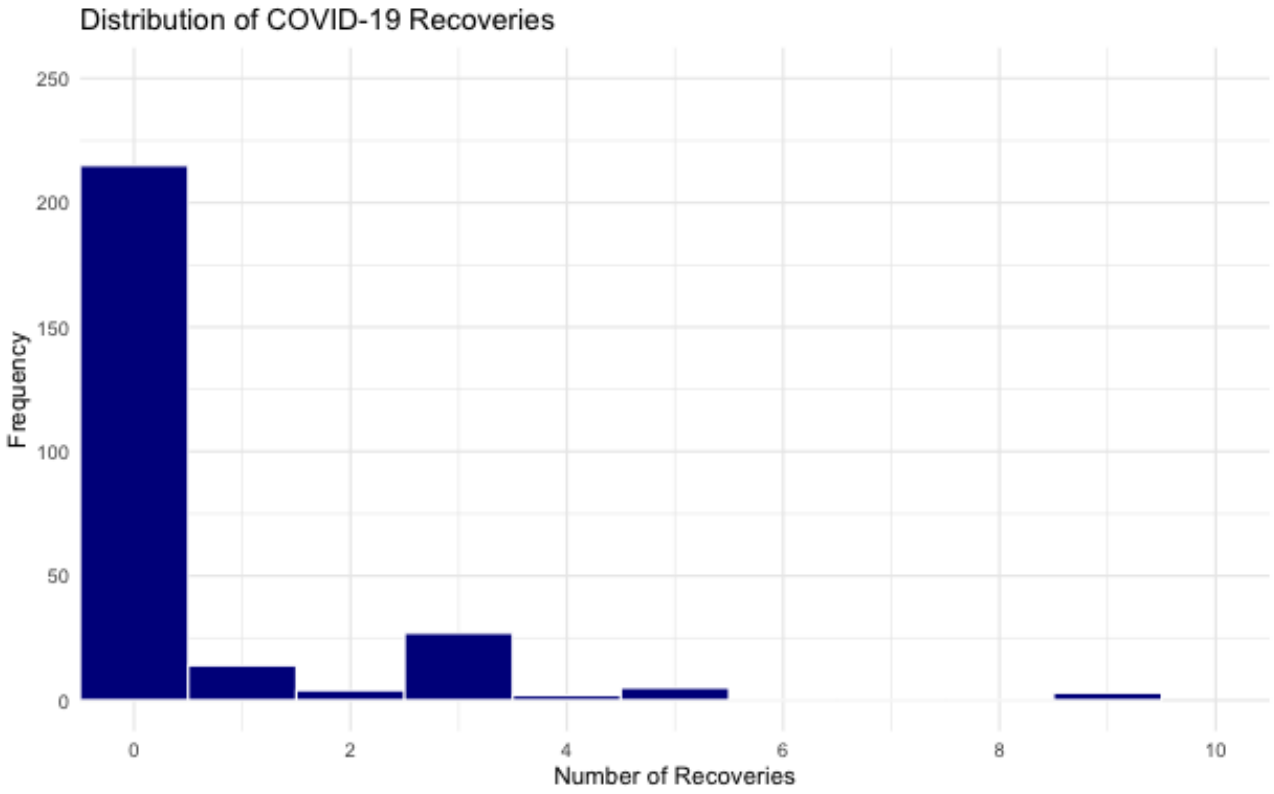
ggplot(covid19cases, aes(x = State.UnionTerritory)) +
  geom_bar(fill = "darkblue") +
  labs(
    title = "Frequency of COVID-19 Reports by State/Territory",
    x = "State/Territory",
    y = "Frequency"
  ) +
  coord_flip() +
  theme_minimal()
```



```
> #Task 2 Using variables ConfirmedIndianNational and ConfirmedForeignNational, create a
pie chart showing the distribution of case severity levels based on total confirmed
cases.
>
> # Calculate total confirmed cases
> covid19cases$TotalConfirmed <- covid19cases$ConfirmedIndianNational +
+   covid19cases$ConfirmedForeignNational
>
> # Categorical Variable breaking severity up into groups
> covid19cases$case_level <- cut(
+   covid19cases$TotalConfirmed,
+   breaks = c(-Inf, 0, 5, 15, Inf),
+   labels = c("No Cases", "Low Cases", "Medium Cases", "High Cases"),
+   right = TRUE)
>
> # Create Severity Table
> case_level <- table(covid19cases$case_level)
>
> # Removes zero cases
> case_level <- case_level[case_level > 0]
>
> ggplot(as.data.frame(case_level),
+        aes(x = "", y = Freq, fill = Var1)) +
+   geom_col(width = 1) +
+   coord_polar(theta = "y") +
+   scale_fill_manual(values = c(
+     "Low Cases" = "lightblue",
+     "Medium Cases" = "orange",
+     "High Cases" = "red"
+   )) +
+   labs(
+     title = "Distribution of COVID-19 Case Severity Levels",
+     fill = "Case Severity"
+  ) +
```



```
> # Task 3 Create a histogram showing the distribution of recovery numbers
>
> ggplot(covid19cases, aes(x = Cured)) +
+   geom_histogram(
+     binwidth = 1,
+     boundary = -0.5,
+     fill = "darkblue",
+     colour = "white"
+   ) +
+   labs(
+     title = "Distribution of COVID-19 Recoveries",
+     x = "Number of Recoveries",
+     y = "Frequency"
+   ) +
+   scale_x_continuous(breaks = seq(0, 10, 2)) +
+   scale_y_continuous(breaks = seq(0, 250, 50)) +
+   coord_cartesian(xlim = c(0, 10), ylim = c(0, 250)) +
+   theme_minimal()
```



```
> # Task 4 Create a line chart showing the trend of total cases over time.
> # Convert Date variable to date format
> covid19cases$Date <- as.Date(covid19cases$Date,
+                              format = "%d-%m-%Y")
>
> # Calculate total confirmed cases
> covid19cases$TotalConfirmed <- covid19cases$ConfirmedIndianNational +
+   covid19cases$ConfirmedForeignNational
>
> # Calculate total confirmed cases for each date
> daily_totals <- aggregate(TotalConfirmed ~ Date,
+                           data = covid19cases,
+                           FUN = sum)
>
> # Display results
> daily_totals
         Date TotalConfirmed
1  2020-01-30              1
2  2020-01-31              1
3  2020-02-01              2
4  2020-02-02              3
5  2020-02-03              3
6  2020-02-04              3
7  2020-02-05              3
8  2020-02-06              3
9  2020-02-07              3
10 2020-02-08              3
11 2020-02-09              3
12 2020-02-10              3
13 2020-02-11              3
14 2020-02-12              3
15 2020-02-13              3
16 2020-02-14              3
17 2020-02-15              3
18 2020-02-16              3
19 2020-02-17              3
20 2020-02-18              3
21 2020-02-19              3
22 2020-02-20              3
23 2020-02-21              3
24 2020-02-22              3
25 2020-02-23              3
26 2020-02-24              3
27 2020-02-25              3
28 2020-02-26              3
29 2020-02-27              3
30 2020-02-28              3
31 2020-02-29              3
32 2020-03-01              3
33 2020-03-02              5
34 2020-03-03              6
35 2020-03-04             28
36 2020-03-05             30
37 2020-03-06             31
38 2020-03-07             34
39 2020-03-08             39
40 2020-03-09             46
41 2020-03-10             58
42 2020-03-11             60
43 2020-03-12             74
44 2020-03-13             81
45 2020-03-14             84
46 2020-03-15            110
47 2020-03-16            114
48 2020-03-17            137
49 2020-03-18            151
50 2020-03-19            173
51 2020-03-20            223
52 2020-03-21            283
>
> # Line Chart
> ggplot(daily_totals, aes(x = Date, y = TotalConfirmed)) +
```

```
+    geom_line(colour = "darkblue", linewidth = 1) +
+    labs(
+      title = "Trend of COVID-19 Confirmed Cases Over Time",
+      x = "Date",
+      y = "Total Confirmed Cases"
+    ) +
+   theme_minimal()
```

